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Coordinates: 5°46′15″S 107°07′16″E

Lion Air Flight 610

Lion Air Flight 610 was a scheduled domestic flight operated by the Indonesian airline Lion Air from Soekarno–Hatta International Airport in Jakarta to Depati Amir Airport in Pangkal Pinang. On 29 October 2018, the Boeing 737 MAX 8 operating the route crashed into the Java Sea 13 minutes after takeoff, killing all 189 passengers and crew.

It was the first major accident involving the new Boeing 737 MAX series aircraft, introduced in 2017, and the deadliest involving the Boeing 737 series, surpassing Air India Express Flight 812 in 2010. It is the deadliest accident in Lion Air's 18-year history, surpassing the 2004 crash in Surakarta that killed 25, and the second deadliest aircraft accident in Indonesia behind Garuda Indonesia Flight 152.

The Indonesian government deployed a search and rescue operation which found debris early the same morning and recovered human remains from a 280 km (150 nmi) wide area. The first victim was identified two days after the crash. The flight data recorder was located on 1 November and recovered for analysis. One member of the volunteer rescue team died during recovery operations. The cockpit voice recorder was found on 14 January 2019.[1]

Preliminary investigations revealed serious flight control problems that traumatized passengers and crew on the aircraft's previous flight, as well as signs of Angle of attack (AoA) sensor and other instrument failures on that and previous flights, tied to a design flaw involving the Maneuvering Characteristics Augmentation System (MCAS) of

Lion Air Flight 610

PK-LQP, the aircraft involved photographed in September 2018, six weeks before the accident

Accident	
Date	29 October 2018
Summary	MCAS design flaw, pilot error, inadequate maintenance
Site	Java Sea, off the north coast of Karawang Regency, Indonesia 5°46′15″S 107°07′16″E
Aircraft	
Aircraft type	Boeing 737 MAX 8
Operator	Lion Air
IATA flight No.	JT610
ICAO flight No.	LNI610
Call sign	Lion Inter 610
Registration	PK-LQP
Flight origin	Soekarno–Hatta International Airport Jakarta, Indonesia

the 737 MAX series. As a result, the United States Federal Aviation Administration and Boeing issued warnings and training advisories to all operators of the 737 MAX series to avoid letting the MCAS cause an abrupt dive similar to the Lion Air flight.

These advisories were not fully implemented, however, and the design issues are suspected to be involved in the Ethiopian Airlines Flight 302 crash on 10 March 2019, leading to the worldwide grounding of all 737 MAX aircraft.

A summary of the October 2019 final report says that "MCAS was the primary cause of the crash, a faulty sensor, inadequate maintenance, poor pilot training and a failure to heed previous problems with the same aircraft were all contributing factors."[2]

Destination	Depati Amir Airport Pangkal Pinang, Indonesia
Occupants	189
Passengers	181
Crew	8
Fatalities	189 (+1 rescue diver)
Survivors	0

[Full screen]

Crash site of Lion Air Flight 610 relative to its origin and destination

Contents

Aircraft
Passengers and crew
- Crew
- Passengers

Flight details and accident
Response
Recovery operations
Investigation
- International assistance
- Previous flight problems
- Early assessment of potential factors
- Flight abnormalities
- November findings
- Preliminary report
- Cockpit voice recorder
- Final report

Aftermath
- Reaction
  - Potential cause

FAA warnings
Pilot concerns
Boeing's defence
Repercussions
Second accident
See also
Notes
References
External links

Aircraft

The aircraft involved was a Boeing 737 MAX 8, registration PK-LQP, line number 7058, powered by two CFM International LEAP engines.[3] The aircraft was leased from China Minsheng Investment Group (CMIG) Aviation Capital and delivered new to Lion Air on 13 August 2018.[4][5] At the time of the accident, the aircraft had flown about 800 hours in service.[6] This was the first accident involving a 737 MAX since the type's entry into service on 22 May 2017,[7] and the deadliest accident involving a Boeing 737.[8]

Passengers and crew

There were 189 people on board the aircraft: 181 passengers (178 adults, 1 child and 2 infants), as well as six cabin crew and two pilots.[9] Officials confirmed that all 189 passengers and crew on board had been killed.[10][11]

Crew

The flight's cockpit crew were captain Bhavye Suneja, a 31 year old Indian national[12] who had flown with the airline for more than seven years and had about 6,028 hours of flight experience (including 5,176 hours on the Boeing 737);[13] and Indonesian co-pilot Harvino,[14][a] who had 5,174 hours of flight experience, 4,286 of them on the Boeing 737. The six flight attendants were also Indonesian.[16]

Passengers

Nationality	Passengers	Crew	Total
Indonesia	180	7	187
Italy	1	0	1
India	0	1	1
Total	**181**	**8**	**189**

Twenty Ministry of Finance employees,[17] 10 Audit Board of Indonesia employees,[18] two auditors from the Finance and Development Inspection Agency,[19] three Ministry of Energy and Mineral Resources employees,[20] three public attorneys,[21] three Indonesian National Police officers,[22] six Bangka Belitung Regional People's Representative Council members,[23] and three judges of Indonesia's High Court and National Court,[24] for a total of 38 civil servants, three police officers and 10 state officials, were among the passengers. There were two confirmed foreigners among those on board: the pilot from India and an Italian citizen, former professional cyclist Andrea Manfredi.[25][26]

Flight details and accident

The aircraft took off from Jakarta on 29 October 2018 at 6:20 a.m. local time (28 October 2018, 11:20 p.m. UTC) and was scheduled to arrive at Depati Amir Airport in Pangkal Pinang at 7:20 a.m.[27] It took off in a westward direction before circling around to a northeast heading, which it held until crashing offshore northeast of Jakarta in waters estimated to be up to 115 ft (35 m) deep.[28][17] The flight crew had requested clearance to return to the Jakarta airport 35 km (19 nmi) into the flight.[29][30] The accident site was located 34 km (18 nmi) off the coast of the island of Java.[31]

Map showing flightpath of Lion Air Flight 610

Communication between air traffic control (ATC) and Flight 610 was suddenly lost at 6:33 a.m.[32] ATC immediately informed authorities about the incident and the Indonesian National Search and Rescue Agency deployed three ships and a helicopter to the area.[33] At 7:30 a.m., the agency received reports that Flight 610 had crashed a few kilometres from an offshore oil platform.[34] Workers on the platform reportedly saw the aircraft crash with a steep nose-down angle.[35] Boats from the platform were immediately deployed and debris from the crashed aircraft was found shortly after.[36]

Response

On 29 October, Indonesia's Transportation Ministry ordered all of the country's airlines to

conduct emergency inspections on their 737 MAX 8 aircraft. The ministry would also launch a special audit on Lion Air to see if there were any problems with its management system.[37] The Transportation Ministry announced that all Indonesian Boeing 737 MAX 8 aircraft were airworthy and were allowed to resume normal operations on 31 October.[38]

A Basarnas spokesperson confirmed to reporters that the aircraft had crashed;[39] Muhammad Syaugi, head of Basarnas, later confirmed that there had been casualties, without specifying a number.[28]

In response to the crash, the Indonesian Transportation Ministry set up crisis centres in Jakarta[40] and Pangkal Pinang.[41][42] Lion Air also offered free flights for the families of the victims to Jakarta. On 30 October, more than 90 relatives were flown to Jakarta for the identification of the victims.[43] CEO of Lion Air Edward Sirait stated that accommodation had been provided for the relatives, and later added that relatives should go to Halim Perdanakusuma International Airport for further information.[44] The Indonesian National Police announced that a trauma healing team would be provided for the relatives of the victims.[45]

Indonesian Minister of Transportation Budi Karya Sumadi giving a press conference on the crash of Flight 610

Indonesia's People's Representative Council announced on 29 October that they would examine the Standard Operating Procedure of Lion Air and the airworthiness of the aircraft. They would also examine the health history of the crew of Flight 610.[46] The Speaker of the People's Representative Council, Bambang Soesatyo, later asked the government to enforce stricter rules for the aviation industry and to audit every airliner in the country.[47] On 1 November, Indonesian Minister of Transportation Budi Karya Sumadi announced that the government would evaluate every low-cost carrier in Indonesia.[48]

As 20 of the passengers were employees of the Indonesian Ministry of Finance, Sri Mulyani, the Indonesian Finance Minister, immediately visited the Indonesian Search and Rescue Agency's office in Jakarta, seeking coordination and further information.[49] She later announced that every employee of her ministry should wear a black ribbon for a week to commemorate the victims.[50] Posthumous awards would be given to the 20 victims and scholarships would be given to the victims' children.[51]

Minister of Health Nila F Moeloek[52] and Minister of Transportation Budi Karya Sumadi visited the relatives of the victims. Indonesian President Joko Widodo, who was attending a conference in Bali during the crash,[53] visited the recovery efforts at the Port of Tanjung Priok the next day.[54]

The Australian Department of Foreign Affairs and Trade announced that its staff would be banned from flying on Lion Air, as well as its subsidiary airlines Batik Air and Wings Air, until the cause of the accident was known.[55] The Indonesian Minister of Transportation, Budi Karya Sumadi, later stated that his ministry would hold talks with the Australian government about the warning.[56][57]

The government-owned social insurance company Jasa Raharja announced that the victims' families would each receive 50 million rupiah (US$3737) in compensation.[58]

In the immediate aftermath of the crash, the Indonesian media were warned by Indonesian Broadcasting Commission for their unethical coverage of the crash. Some media were accused of pushing the family members to answer unethical questions.[59] Chairman of Indonesia's Alliance of Independent Journalists (AJI), Abdul Manan, stated that images of debris were broadcast repeatedly and inappropriately. This, in turn, traumatized relatives of the victims. In response, some Indonesians voiced their anger and disappointment on social media.[60]

On 31 October, Transportation Minister Budi Karya Sumadi temporarily suspended Lion Air's technical director Muhammad Arif from his duties based on the crash investigation. Budi said the ministry had also suspended an unspecified number of Lion Air technicians who had cleared the aircraft for its final flight.[61]

Recovery operations

Indonesia's National Search and Rescue Agency (Basarnas) launched a search and rescue operation, with assistance from the Indonesian Air Force,[62] the Indonesian Navy,[63] and the Republic of Singapore Navy. Basarnas dispatched about 150 people in boats and helicopters to the site of the accident.[28] Civilian vessels also responded to the reports of a downed aircraft, and the crew of a tugboat reported to authorities in Tanjung Priok that they had witnessed an aircraft crash at 6:45 a.m. and located debris in the water at 7:15 a.m.[27][39] The Indonesian Agency for Assessment and Application for Technology deployed the research ship *Baruna Jaya*, which had been previously used in the search for Adam Air Flight 574 and Indonesia AirAsia Flight 8501.[64]

Officials from the Indonesian National Search and Rescue Agency announced that the search and rescue operation would be conducted for seven days and be extended by three days if needed. A command centre was set up in Tanjung Priok.[65]

On 29 October, authorities said that all on board were presumed dead and that the first human remains had been recovered.[66] Divers had located fragments of the aircraft's fuselage and assorted debris, but had yet to find the onboard flight recorders.[66] An official from the Indonesian National Armed Forces suggested that most of the victims were still

inside the fuselage, as in the days following the crash rescue personnel only managed to recover a small number of body parts.[67] Officials stated that bad visibility and strong sea current hampered the search and rescue effort.[68]

On the same day, the Indonesian National Search and Rescue Agency published the area of the search and rescue operation. It was divided into two main areas. The first was a 115 km (62 nmi) underwater search area while the second was a "visual search" area, 124 nautical miles (230 km; 143 mi) wide.[69]

On 30 October, the search area was divided into 13 sectors. The search area was widened to 280 km (150 nmi),[70] reaching as far as Indramayu to the east. Approximately 850 personnel from the Indonesian National Search and Rescue Agency, Indonesian National Armed Forces and volunteers participated in the operation.[71] At least 13 bodies were retrieved from the crash site.[72] Indonesian officials confirmed that faint pings had been heard in the search area.[73]

On 31 October, it was reported that acoustic "pings" had been detected, no further than 3 km (1.6 nmi) from the group of eight current search points, which were possibly from one or both of the underwater locator beacons (ULBs) attached to the aircraft's flight recorders.[74]

The first victim was identified on 31 October.[75] At the time, more than a dozen body parts had been found by authorities. Some of the parts had drifted more than 5 km (2.7 nmi) in the sea current. Police also reported that 152 DNA samples had been collected from the victims' relatives.[75] Hundreds of pieces of the aircraft had also been recovered; all of them were transported to Tanjung Priok, Jakarta.[76] Authorities stated that the search area for dead bodies and debris would be focused in the sea off Karawang Regency, a coastal area of Java close to the crash site, as analysis showed that the sea currents in the area would bring debris to the south. A command centre was set up in Tanjung Pakis, Karawang to oversee the salvage effort.[76]

On the same day, authorities widened the search area from 19 to 28 km (10 to 15 nmi). In all, 39 ships (including 4 equipped with sonar) and 50 divers were deployed to the search area.[77] The Indonesian National Police announced that 651 personnel had joined and assisted in the search and rescue operation.[78] Officials stated that the operation, starting from 31 October, would focus on finding the fuselage of the aircraft and the flight recorders.[79]

The joint search and rescue team announced on 31 October that at least three objects, one of which was suspected to be one of the aircraft's wings, were found in the search area.[80] Officials confirmed that "pings" from the aircraft's ULBs were also heard near the area.[81]

On 1 November, searchers announced that they had found Flight 610's flight data recorder (FDR), which was located at a depth of 32 m (105 ft). The cockpit voice recorder (CVR),

however, was reported as not yet found.[82] According to a transport safety official, quoted by Reuters news agency, authorities did not initially know for certain whether the "crash survivable memory unit" was from the flight data recorder or cockpit voice recorder, as portions of it were missing.[83] Haryo Satmiko, deputy chief of Indonesia's National Transportation Safety Committee (NTSC) (Indonesian: *Komite Nasional Keselamatan Transportasi*), the body investigating the crash, told journalists that the device's poor condition was evidence of the "extraordinary impact" of the crash, which had separated the memory unit from its housing.[84][85] Despite the damage, investigators were able to recover data from the aircraft's most recent 19 flights spanning 69 hours, and planned to begin analysis on 5 November.[85]

On 2 November, the joint search and rescue team deployed more than 850 personnel and 45 vessels to the crash site. The aerial search area was widened to 350 km (190 nmi) and the "underwater search area" was widened to 500 km (270 nmi).[86] Joint search and rescue team announced that some engine parts were found in the search area.[87] One of the aircraft's landing gears was recovered in the afternoon.[88] Meanwhile, the Disaster Victim Identification team stated that at least 250 body parts had been recovered from the crash site.[89]

On 3 November, it was reported that a volunteer Indonesian rescue diver had died during the search, on the afternoon of 2 November.[90] It is believed he died from decompression sickness.[91]

A second landing gear and both of the aircraft's engines were recovered by search and rescue personnel, and the main body of the aircraft had been located. The main wreckage of the aircraft was located 13.9 km (7.5 nmi) from the coast of Tanjung Pakis and was about 200 m (220 yd) from the location where the FDR was discovered. Divers were immediately dispatched to the area. Faint "pings" from the ULB attached to the aircraft's CVR were also heard.[92][93][94]

On 4 November, nearly 1,400 personnel, including 175 divers, were dispatched to the crash site. 69 ships, 5 helicopters and 30 ambulances were also dispatched.[95] The Head of the Indonesian National Search and Rescue Agency Muhammad Syaugi announced that the search and rescue operation would be extended for another three days.[96]

On 10 November the NSRA ended its search for victims,[97] but on 22 November were continuing to search intensively for the CVR.[98]

On 23 November, investigators concluded the victim identification process. Out of 189 people on board, 125 (89 men and 36 women) were identified, including the two foreigners. Another 64 bodies are still unaccounted for.[99]

On 14 December, it was reported that Lion Air had paid US$2.8 million for a second attempt

to search for the CVR, with a specialized boat being brought in to assist in the search, expected to last 10 days.[100] The attempt began on 20 December,[101] but ended on 3 January 2019 without success.

The NTSC announced plans to launch their own search, and were negotiating with the Indonesian Navy to borrow one of their ships.[102] On 8 January 2019, they announced they would be resuming the search for the CVR (using their own funding).[103] The CVR was finally found and recovered by the Indonesian Navy near the crash site, at a depth of 30 m (98 ft) but covered by mud that was 8 m (26 ft) thick,[104] on 14 January 2019, more than two months after the accident.[1][105]

Investigation

International assistance

A Boeing technician and engineering team, and a team from the US National Transportation Safety Board arrived on 31 October to help with the investigation being conducted by the NTSC. Personnel from the US Federal Aviation Administration (FAA) and engine manufacturer GE Aviation were also sent to Indonesia. A team from Singapore, that had already arrived on the night of 29 October, was to provide assistance in recovering the aircraft's flight recorders. The Australian Transport Safety Bureau sent two of its personnel to assist with the downloading process of the FDR.[106]

Previous flight problems

The aircraft was used on a flight from Ngurah Rai International Airport, Bali to Soekarno-Hatta International Airport, Jakarta the night before the crash. Detailed reports from that flight revealed that the aircraft had suffered a serious incident, which left many passengers traumatized. Passengers in the cabin reported heavy shaking and a smell of burnt rubber inside the cabin. At one point, the aircraft had dropped more than 61 m (200 ft) in a few seconds.[107] The seat belt sign was never turned off from takeoff to landing. A recording of air traffic control communications indicated that the pilot had called a "pan-pan."[108] The crew later decided to cancel the pan-pan and continue the flight to Jakarta.[107]

The aircraft's maintenance logbook revealed that the aircraft suffered an unspecified navigation failure on the captain's side, while the First Officer's side was reported to be in good condition.[109]

Passengers recounted that the aircraft had suffered an engine problem and were told not to board it as engineers tried to fix the problem. While the aircraft was en route to Jakarta, it had problems maintaining a constant altitude, with passengers stating that it was like "a

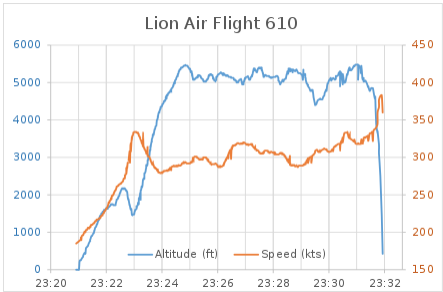
roller-coaster ride."[110] The chief executive officer of Lion Air, Edward Sirait, said the aircraft had a "technical issue" on Sunday night, but this had been addressed in accordance with maintenance manuals issued by the manufacturer. Engineers had declared that the aircraft was ready for takeoff on the morning of the accident.[111][112] A later report claimed that a third pilot was on the flight to Jakarta and told the crew to cut power to the stabilizer trim motors which fixed the problem. This method is a standard memory item in the 737 checklist.[113] Subsequently, the National Transportation Safety Committee confirmed the presence of an off-duty Boeing 737 MAX 8 qualified pilot in the cockpit but did not confirm the role of the pilot in fixing the problem, and denied that there was any recording of the previous flight in the CVR of Lion Air Flight 610.[114]

Early assessment of potential factors

Indonesian aviation expert Gerry Soejatman speculated that the pitot tubes, used in the airspeed indication system, may have played a role in the crash. They have contributed to previous similar crashes.[115][116]

The Indonesian Meteorology, Climatology, and Geophysical Agency (BMKG) reported that the weather was clear around the time of the crash, with winds at 9.3 km/h (5 kn) from the northwest. Visibility was good with no cumulonimbus clouds.[117] Police Hospital Chief Musyafak said that an examination of the body parts indicated that it was unlikely that there had been an explosion or fire on board the aircraft.[118]

Flight abnormalities



Flight JT610 Altitude and Speed

Aviation experts noted that there were some abnormalities in the altitude and the airspeed of Flight 610. Just three minutes into the flight, the captain asked the controller for permission to return to the airport as there were flight control problems.[119] About eight minutes into the flight, data transmitted automatically by the aircraft showed it had descended to about 1,500 m (5,000 ft) but its altitude continued to fluctuate. The mean value of the airspeed data transmitted by Flight 610 was around 560 km/h (300 kn), which was considered by experts to be unusual, as typically aircraft at altitudes lower than 3,000 m (10,000 ft) are restricted to an airspeed of 460 km/h (250 kn). Ten minutes into the flight, the data recorded the aircraft dropping by more than 910 m (3,000 ft). The last recorded altitude of the aircraft was 760 m (2,500 ft).[120][121]

November findings

On 5 November, the NTSC announced that Flight 610 was still intact when it crashed into the sea at high speed, citing the relatively small size of the pieces of debris. The impact was so powerful that the strongest part of the aircraft was obliterated.[122] The NTSC also stated that the engines of Flight 610 were still running when it crashed into the sea,[123] indicated by the high RPM. Further examination of the aircraft's instruments revealed that one of the aircraft's airspeed indicators had malfunctioned for its last four flights, including the flight to Denpasar.[124]

On 7 November, the NTSC confirmed that there had been problems with Flight 610's angle of attack (AoA) sensors. Thinking that it would fix the problem, the engineers in Bali then replaced one of the aircraft's AoA sensors, but the problem persisted on the penultimate flight, from Denpasar to Jakarta. Just minutes after takeoff, the aircraft abruptly dived. The crew of that flight, however, had managed to control the aircraft and decided to fly at a lower than normal altitude. They then managed to land the aircraft safely and recorded a twenty-degree difference between the readings of the left AoA sensor and the right sensor.[125] NTSC chief Soerjanto Tjahjono told the press that future reporting or actions, enacted to prevent similar problems on similar aircraft, would be decided by Boeing and U.S. aviation authorities.[126]

On 28 November, Indonesia investigators said the Lion Air jet was not airworthy on the flight before the crash. Several relatives of the crash victims have already filed lawsuits against Boeing.[127]

Preliminary report

On 28 November, the Indonesian National Transportation Safety Committee (NTSC) released its preliminary accident investigation report.[13] After airspeed and altitude problems, an AoA sensor was replaced and tested two days earlier on the accident aircraft.[128] Erroneous airspeed indications were still present on the subsequent flight on 28 October, which experienced automatic nose down trim.[128] The runaway stabilizer non-normal checklist was run, the electric stabilizer trim was turned off, and the flight continued with manual trim; the issues were reported after landing.[128] Shortly after takeoff on 29 October, issues involving altitude and airspeed continued due to erroneous AoA data and commanded automatic nose-down trim via the Maneuvering Characteristics Augmentation System (MCAS).[128] The flight crew repeatedly commanded nose-up trim over the final ten minutes of the flight.[128] The preliminary report did not state whether the runaway stabilizer trim procedure was run or whether the electric stabilizer trim switches were cut out on the accident flight.[128]

Leeham News, which principally covers Airbus, Boeing, Bombardier and Embraer issues,[129] reported that the crew did not have a clear knowledge that trim runaway checklist will disengage MCAS.[130]

Boeing pointed to the successful troubleshooting conducted on 28 October as evidence that the MCAS did not change runaway stabilizer procedures,[131] and emphasised the longstanding existence of procedures to cancel MCAS nose-down commands.[132]

Cockpit voice recorder

On 21 January 2019, the NTSC announced that it would not release transcript from the cockpit voice recorder (CVR) found a week earlier until the final report was released.[133] However, following the Ethiopian Airlines Flight 302 crash on 10 March 2019 which meant the data from the CVR of the Lion Air Flight 610 were shared among the investigators, media citing anonymous sources reported that CVR recorded the pilots mentioning several problems, trying to climb, and checking the quick reference handbook for a solution.[134] NTSC denied these claims but stated the pilots began to panic at the end of the flight.[135]

Final report

The NTSC released its final report into the accident on 25 October 2019.[136][137] According to a summary of the report, the contributing factors and safety recommendations are as follows:

1. Boeing:

    a) MCAS: The assumption made by Boeing (approved by FAA) on pilots response were incorrect. Reliance of MCAS on single AOA sensor was deemed appropriate (based on the assumption above), but this decision of using single AOA sensor caused it to be vulnerable to erroneous sensor input. The absence of MCAS guidance (in pilot's manual and training material) made it difficult (impossible) for crew to properly respond to uncommanded trim caused by MCAS.

    b) The AOA DISAGREE alert was not enabled, as the result it did not appear and not documented by the (DPS-CGK) flight crew, and therefore not identified by maintenance.

2. Lion Air, Batam Aero Technic (Lion Air Group Indonesia) and Xtra Aerospace USA:

    a) The replacement of the AOA sensor installed on the airplane had been mis-calibrated by the repair shop (Xtra Aerospace,Florida). On both the previous (DPS-CGK) and the accident flight the left AOA sensor was reading 21 degrees too high since the beginning of the takeoff. The US NTSB and Boeing conducted a test with a 33 degrees biased AOA sensor and concluded a Mis-calibrated sensor would be detected by the airplane's SMYD computer during installation. Xtra's calibration device's selector was in the REL instead of ABS, the mis-calibration/bias of 21 degrees too high was not detected as the result. After Lion 610 crash FAA did an audit on Xtra at the end of 2018 and there were numerous findings. After the Final Report of JT610 is released the FAA revoked Xtra Aerospace certificate, preventing the company from doing business.

b) Post installation test carried out by Batam Aero Technic (Lion Air Group) engineer in DPS was not done properly. Had the procedure was done correctly the airplane's SMYD (Stall Management Yaw Damper) computer would have had detected the mis-calibration and "AOA SENSR INVALID" message will appear. The engineer in Denpasar provided the investigation several photos including of the Captain's PFD that was claimed to be taken after the AOA sensor replacement and of the SMYD during the installation test. However, the time shown on the Captain's PFD was the time before arrival of AOA sensor spare part and the investigation confirmed that the SMYD photos were not of the accident aircraft. On both the previous (DPS-CGK) and the accident flight the left AOA sensor was reading 21 degrees too high since the beginning of the takeoff.

c) The flight crew of the previous(DPS-CGK) flight did not mention the continuous activation of stick shaker (which is caused by the AOA sensor), and the use of Runaway stabiliser NNC. this lack of information to maintenance crew lead into the maintenance crew being mislead that the faulty AOA was not part of the problem, and the accident flight crew did not consider they could reoccur.

d) The pilots did not respond appropriately during the final (accident flight):

There was a lengthy delay for the UNRELIABLE AIRSPEED Non-Normal checklist. The captain nor the FO did not do it from memory immediately despite it is an industry standard that all NNCs classified as memory items should be executed without delay from memory. The FO had difficulties finding the UNRELIABLE AIRSPEED non-normal checklist on the QRH.

The pilots did not declare emergency (MAYDAY) or urgency (PANPAN) to ATC, as the result they were not receiving priority and instead receive normal vectoring which actually added to their already heavy workload.

The breakdown of CRM (Crew Resource Management): when the captain transferred the controls to the FO he did not communicate effectively that he needed to counteract multiple trim down (caused by MCAS). Therefore the FO could not anticipate the requirement to repeatedly trim up countering the MCAS. The manual flying skill of the FO have been identified during training.

Furthermore the KNKT report stated that there have been 31 pages missing from the airplane's log book (AFML), Lion Air are to improve the duration and content of Safety Management System (SMS) training including the identification of hazard (this is the result of the previous pilot failure to report the continuous stick shaker and stab trim runaway NNC), and the Indonesian DGCA to improve the oversight towards airlines and maintenance organisation.[138][139]

Aftermath

Reaction

Potential cause

The CFM International LEAP engines of the 737 MAX have a higher bypass ratio and have a larger nacelle than the engines of previous Boeing 737 models, so the engines are placed higher and further forward in relation to the wing than on previous models. This destabilises the aircraft pitch at higher angles of attack (AoA). To deal with this, Boeing designed a Maneuvering Characteristics Augmentation System (MCAS) for the 737 MAX series.[140] Former Boeing engineers expressed the opinion that a nose down command triggered by a sensor single point of failure is a design flaw if the crew is not prepared, and the FAA was evaluating a fix of the possible flaw and investigating whether the pilots' transition training is adequate.[141] A malfunction in the AoA sensors could lead the MCAS system to believe that the aircraft is stalling when it isn't, causing it to dip the aircraft's nose to recover from a non-existent stall even in level flight.[142]

FAA warnings

On 7 November, on the basis of preliminary information gathered in the investigation of the Lion Air accident, the Federal Aviation Administration (FAA) issued an emergency Airworthiness Directive (AD) requiring that amended operating limitations and procedures relating to erroneous data from an AoA sensor be inserted into the aircraft flight manual of each 737 MAX aircraft,[143][144] and urged all airlines operating Boeing 737 MAX 8s to heed the warnings.[145]

On 25 October 2019, after the release of the Final Report by NTSC, FAA revoked the repair certification of Florida-based Xtra Aerospace LLC, which fixed a AOA sensor suspected of contributing to the crash.[146]

Pilot concerns

During difference training, pilots of American Airlines and Southwest Airlines converting from earlier Boeing 737 Next Generation models to the 737 MAX were not informed of the MCAS linked to the fatal crash, leaving them concerned that they were possibly untrained with respect to other differences.[147] In November 2018, *Aviation Week* reviewed the 737 MAX flight crew operations manual and found that it did not mention the MCAS.[148] American Airlines' Allied Pilots Association and Southwest Airlines Pilots' Association were also caught unaware.[149] *The Wall Street Journal* reported that Boeing had "decided against disclosing more details to cockpit crews due to concerns about inundating average pilots with too much information."[150]

On 15 November, the US Air Line Pilots Association (ALPA), the largest pilots' union, representing 61,000 pilots, urged the FAA and NTSB to ensure pilots receive all relevant information addressing a "potential, significant aviation system safety deficiency."[151] ALPA's United Airlines branch, in line with its management, disagrees as the 737 pilot

manual includes a standard procedure to shut down the flight-control behavior, and dismissed the MCAS implication in the accident as "speculation" based on the Boeing safety-warning bulletin and the follow-on FAA airworthiness directive.[152]

Boeing's defence

In an internal message on 19 November 2018, Boeing CEO Dennis Muilenburg defended the Flight Crew Operations Manual as describing the relevant function of MCAS.[153] On 20 November, Boeing was to hold a conference call with 737 Max operators to detail the new MCAS not present in the Next Generation models.[154] The conference call was cancelled later, to be replaced by a series of regional calls to allow more questions.[155]

On 25 October 2019, after the release of the final investigation report by NTSC, Boeing President and CEO Dennis Muilenburg responded:

> "We are addressing the KNKT's safety recommendations, and taking actions to enhance the safety of the 737 MAX to prevent the flight control conditions that occurred in this accident from ever happening again. Safety is an enduring value for everyone at Boeing and the safety of the flying public, our customers, and the crews aboard our airplanes is always our top priority. We value our long-standing partnership with Lion Air and we look forward to continuing to work together in the future."[156]

Repercussions

Lion Air co-founder and former CEO Rusdi Kirana, currently the Indonesian ambassador to Malaysia, reportedly considered cancelling Lion Air's outstanding 190 Boeing aircraft orders – worth some $22 billion at list prices – over what he viewed as an attempt by Boeing to blame Lion Air for the crash.[132]

On 31 December, the family of the first officer filed a lawsuit against Boeing, claiming negligence. The lawsuit also claimed that the aircraft's sensors provided inaccurate flight data, causing its anti-stall system to improperly engage, as well as Boeing not providing proper instructions to pilots about how to handle the situation.[157]

In March 2019, victims' families reported irregularities, saying that Lion Air pressured them into signing away their rights to seek legal recourse for under-compensation.[158][159]

Second accident

On 10 March 2019, another 737 MAX 8, operated by Ethiopian Airlines (registration ET-

AVJ), crashed shortly after takeoff from Addis Ababa; all 157 people on board perished. This raised further concerns about the safety of the 737 MAX and culminated in all 737 MAX aircraft (regardless of variant) being grounded worldwide for an indefinite amount of time.[160]

See also

- Ethiopian Airlines Flight 302
- Boeing 737 MAX groundings
- Air France Flight 447
- List of accidents and incidents involving the Boeing 737
- List of aviation accidents and incidents in Indonesia
- List of accidents and incidents involving commercial aircraft
- List of aircraft accidents and incidents resulting in at least 50 fatalities

Notes

a. Harvino, like some Indonesians, only had a single name[15] - See Indonesian name

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